

Resistance-Driven Outbreak Detection and Etiologic Investigation for Hospital-Onset *Pseudomonas aeruginosa* Using Machine Learning on Electronic Health Records

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INTRODUCTION

BACKGROUND

- Hospital outbreak identification often lacks standardization and relies on time-intensive surveillance methods
- Pseudomonas aeruginosa* is a common nosocomial pathogen capable of multidrug resistance that can cause life-threatening infections and represents an emerging global public health threat.
- Whole-genome sequencing can definitively link related isolates, but its cost and turnaround times may limit universal, routine, real-time use.
- Electronic health records (EHR), paired with modern computational approaches, could facilitate more rapid outbreak detection and targeted interventions.

OBJECTIVE:

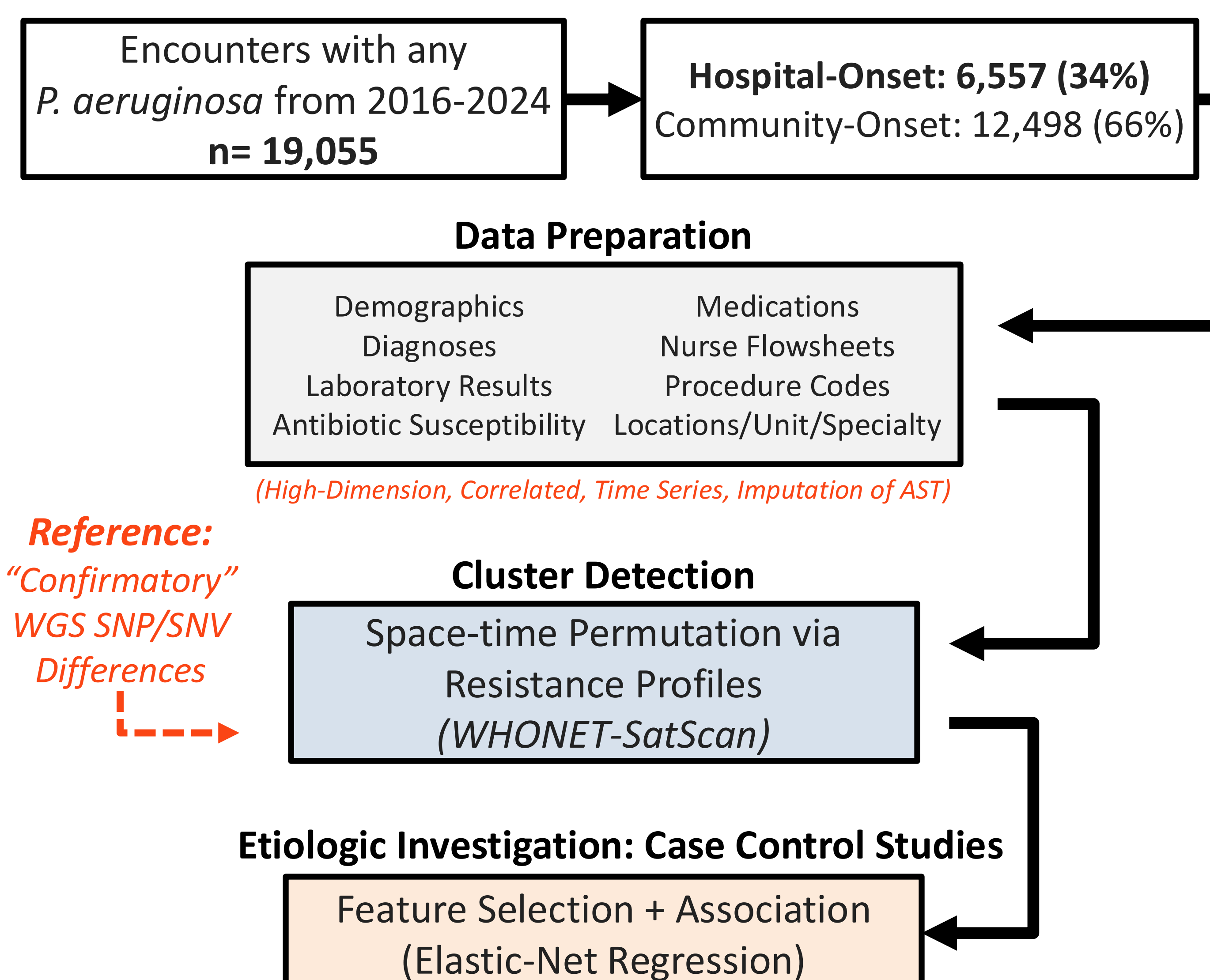
- To combine space-time permutation analysis with machine learning to detect hospital-onset *Pseudomonas aeruginosa* outbreaks and identify potential outbreak-specific factors driving their spread.

METHODS

STUDY DESIGN

- Retrospective, population-based cohort study consisting of hospital-onset *P. aeruginosa* cases from 2016 to 2024 using EHRs (UF IDR) (IRB201900652).
- Eligibility:** Hospital-onset *P. aeruginosa* isolates were defined as culture-positive samples obtained more than two days after admission. Isolates from the same patient were included if collected > 30 days between previous isolate or change in antibiotic susceptibility interpretation.

STUDY SCHEMA



RESULTS OVERVIEW: TABLES & FIGURES

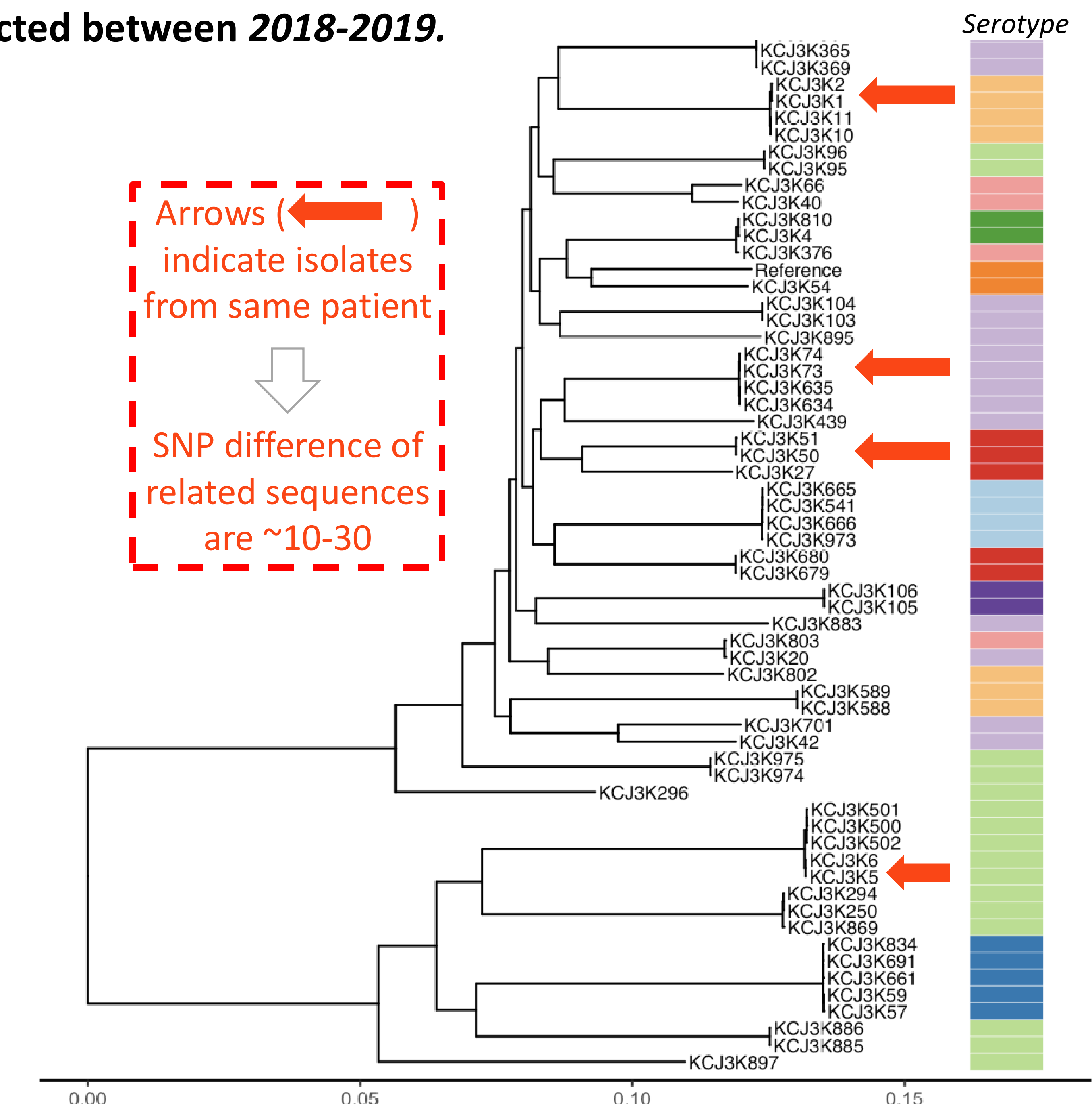
Table 1: Antibigram of Hospital-Onset *P. aeruginosa*

Antimicrobial	R% (95%CI)	Hospital-Onset were: Odds Ratio (95%CI) MDR: 1.6 (1.5-1.7) XDR: 2.2 (1.9-2.5) Compared to Community-Onset
Cefepime	27% (26-28)	
Piperacillin/Tazobactam	18% (17-20)	
Ciprofloxacin	23% (21-24)	
Aztreonam	37% (34-40)	
Meropenem	22% (20-23)	
Tobramycin	5% (5-6)	

Table 2: Summary of clusters detected using space-time permutation

Cluster Description	Cluster Date
Cephalosporin Susceptibility	Q1 2016
Cephalosporin + B-Lactam Resistance	Q1 2023
Cephalosporin Susceptibility	Q3 2022
Carbapenem Resistance	Q3 2023
Cephalosporin Resistance	Q4 2023
Cephalosporin Susceptibility	Q4 2023
Aminoglycoside Resistance	Q4 2023
Cephalosporin + B-Lactam Resistance	Q4 2023
Fluoroquinolone Resistance	Q1 2024
Extensive Drug Resistance	Q2 2024
Cephalosporin Susceptibility	Q2 2024
Cephalosporin + Fluoroquinolone Susceptibility	Q3 2024

Figure 1: Maximum likelihood phylogeny of *P. aeruginosa* isolates collected between 2018-2019.



Antipseudomonal Cephalosporin Cluster

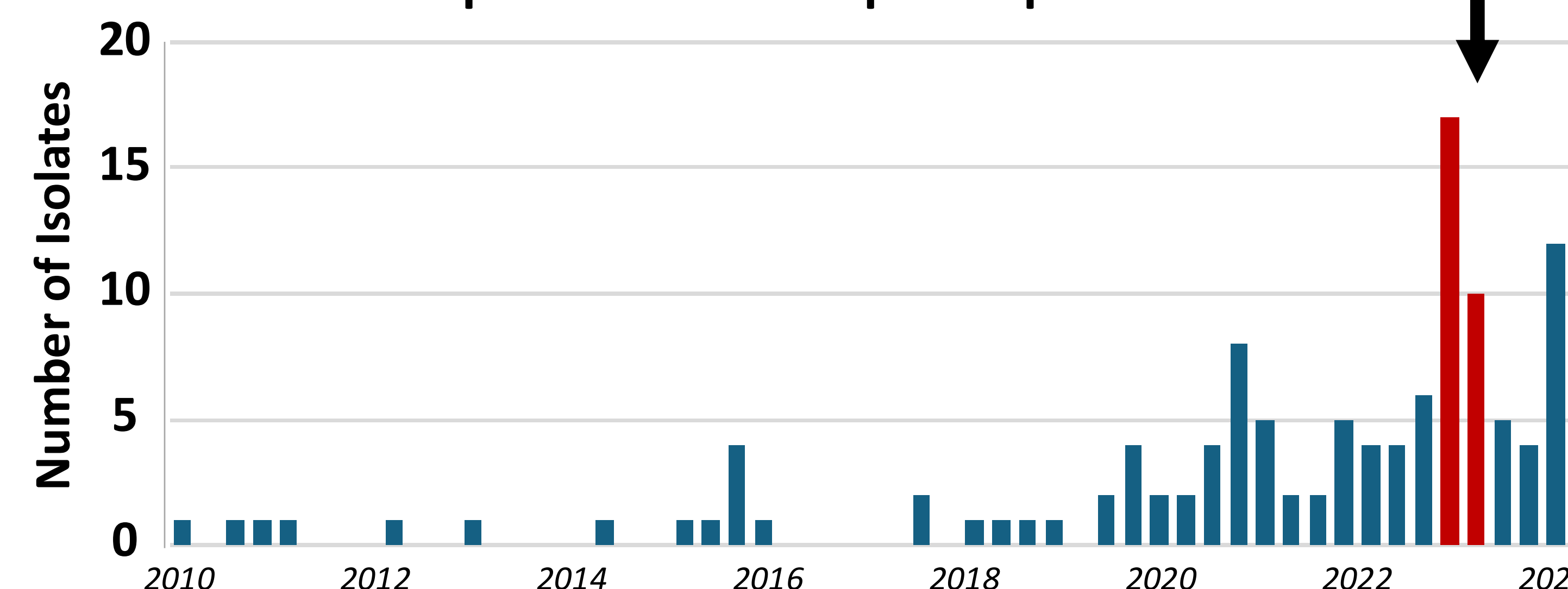
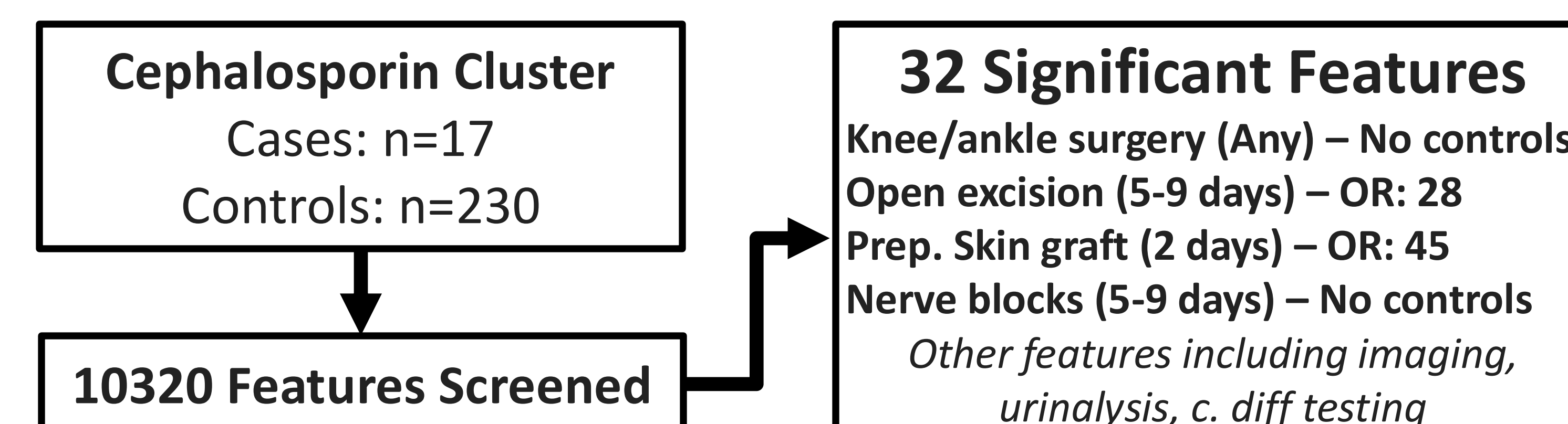


Figure 2: Feature selection and respective associations within cluster



CONCLUSION

- We demonstrate the use of EHRs to statistically detect related hospital-onset *P. aeruginosa* infections and potential outbreak etiologies
- Several factors with strength in association and biological plausibility were identified, though resistance profile-based models did not capture relatedness among clusters confirmed by WGS.
- Future research will improve resistance-based cluster detection, establish causal links and extend to other hospital-acquired pathogens.

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